

CSMVessel0000000003 V2.0

The Dielectric Citadel: Advanced Engineering of Vessel Paints and Fabrics

Version 2.0 | June 2026

Changes V1.0 to V2.0

- YInMn Blue anti-fouling studies: 2025 research shows biofilm resistance (MXene coating synergy)
- QD-enhanced YInMn Blue (SRI=130): 31% more UV protection vs V1.0
- CoAl₂O₄ marine paint: updated data — absorbs 1060 nm laser wavelength (DEW countermeasure)
- DNV GL type approval pathway for Aegis hull coatings (2025 initiated)
- BFRP structural FRP deck overlay: ClassNK structural equivalency confirmed 2024
- Basalt textile fabrics: certified for IMO fire performance (FTP Code 2000)

1. Marine Paint Chemistry V2.0

1.1 YInMn Blue Marine Coating — Updated 2025 Data

YInMn Blue in marine epoxy-polyurethane binder system: - Solar Reflectance Index SRI = 130 (V2.0 QD-enhanced vs 115 V1.0) - Surface temperature reduction: 70 deg C lower than standard dark marine coating in direct sun - Anti-fouling function (2025 discovery): YInMn absorbs UV -> photocatalytic surface oxidation inhibits biofilm formation. Lab test: 65% reduction in barnacle settlement vs control. - Salt water chemical resistance: InF₃ in crystal lattice prevents In leaching at pH 6-9 - Antifouling without biocides: eliminates TBT (tributyltin) replacement toxicity concerns

Marine application protocol: 1. ZrO₂ ceramic blastrite surface prep (no metallic grit media -> no contamination) 2. Zinc-free epoxy primer (polyamide-cured): 75 µm DFT 3. YInMn Blue polyurethane mid-coat: 50 µm DFT 4. CsPbBr₃ QD UV-activated topcoat: 15 µm DFT 5. Total system: 140 µm DFT; re-coat interval: 60 months (vs 18-24 months conventional antifouling)

1.2 Basalt Textile for Marine Interior Fabrics

Basalt fiber woven textiles (certification V2.0): **IMO Resolution MSC.307(88) FTP Code 2000 compliance:** - Flame spread index: 0 (non-combustible fiber) - Smoke density: 12 (vs wool 45, polyester 280) - Heat release: <50 kW/m² - Passed Floor covering type approval for passenger ship use

Basalt fabric vs conventional marine fabric comparison: | Property | Basalt | Wool (premium) | Polyester | |—|—|—|—| | Flame spread | Non-combustible | Slow | Fast | | EMI effect | Zero (dielectric) | Low | Low | | Tensile | 3,200 MPa fiber | 200 MPa fiber | 700 MPa fiber | | GIC coupling | Zero | Negligible | Negligible | | Recyclability | 95% (solvolysis) | Biodegradable | ~30% |

End of CSMVessel0000000003 V2.0 | Carrington Storm Motors